

Def. Let X, Y be Metric Spaces, and $f_n: X \rightarrow Y$ be function defined for each $n \in \mathbb{N}$.
 For each $x \in X$, define $f(x) \stackrel{\text{def}}{=} \lim_{n \rightarrow \infty} f_n(x)$ if the limit exists.

This f is called limit function of $\{f_n\}$.

Def. Suppose that a sequence of functions $\{f_n\}$ on X into Y satisfies that; the limit function f is defined on X and

For all $\epsilon > 0$, there exists $N \in \mathbb{N}$ s.t. $n \geq N \Rightarrow d_Y(f_n(x), f(x)) \leq \epsilon, \forall x \in X$.

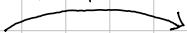
Then, we say that $\{f_n\}$ converges uniformly on X to f . Denote $f_n \rightrightarrows f$.

Note: $\forall x \in X, d_Y(f_n(x), f(x)) \leq \epsilon \Leftrightarrow \sup_{x \in X} d_Y(f_n(x), f(x)) \leq \epsilon$. That is,

$$f_n \rightrightarrows f \text{ on } X \Leftrightarrow \lim_{n \rightarrow \infty} \sup_{x \in X} d_Y(f_n(x), f(x)) = 0.$$

Thm. Let $\{f_n\}$ be functions defined on any Metric Space X into Complete Metric space Y . Then,

In General Metric Space



$\{f_n\}$ converges uniformly on X to f if and only if $\forall \epsilon > 0, \exists N \in \mathbb{N}$ s.t. $m, n \geq N \Rightarrow \sup_{x \in X} d_Y(f_m(x), f_n(x)) \leq \epsilon$

In Complete Metric space

Proof. If $f_n \rightrightarrows f$, then,

$$\sup_{x \in X} d_Y(f_n(x), f_m(x)) \leq \sup_{x \in X} d_Y(f_n(x), f(x)) + \sup_{x \in X} d_Y(f(x), f_m(x)) \rightarrow 0$$

Give the result. (가장 먼저 보자면, Supremum의 성질과 triangle inequality가 한 가지 큰 몫을 차지한다.)

Conversely, since Cauchy sequence in the Complete Metric space converges, the limit function f exists.

$$\text{Now, } \lim_{m \rightarrow \infty} \sup_{x \in X} d_Y(f_m(x), f_n(x)) = \sup_{x \in X} d_Y(f(x), f_n(x))$$

Gives the result. (More specifically, let $n \in \mathbb{N}$ fixed. Observe that

$$\left| \sup_{x \in X} d_Y(f_m(x), f_n(x)) - \sup_{x \in X} d_Y(f(x), f_n(x)) \right|$$

d_Y Continuous



$$\sup_{x \in X} d_Y(f(x), f_n(x)) = \sup_{x \in X} d_Y(\lim_{m \rightarrow \infty} f_m(x), f_n(x)) = \sup_{x \in X} \lim_{m \rightarrow \infty} d_Y(f_m(x), f_n(x)) = \lim_{m \rightarrow \infty} \sup_{x \in X} d_Y(f_m(x), f_n(x))$$

$$\left| \sup_{x \in X} \lim_{m \rightarrow \infty} d_Y(f_m(x), f_n(x)) - \sup_{x \in X} d_Y(f_m(x), f_n(x)) \right| \leq \sup_{x \in X} \left| \lim_{m \rightarrow \infty} d_Y(f_m(x), f_n(x)) - d_Y(f_m(x), f_n(x)) \right| \rightarrow 0$$

Gives (4).

Uniform Convergence and Continuity.

Thm. Suppose that $f_n, f: X \rightarrow Y$ are functions where X is Metric, Y is complete Metric sp.

Let $x \in X$ be a limit point of X .

If for each $n \in \mathbb{N}$, the limit $\lim_{t \rightarrow x} f_n(t)$ exists and $f_n \rightarrow f$ on X ,

then
$$\lim_{t \rightarrow x} \lim_{n \rightarrow \infty} f_n(t) = \lim_{n \rightarrow \infty} \lim_{t \rightarrow x} f_n(t)$$

Proof. Given $\epsilon > 0$, there exists $N \in \mathbb{N}$ s.t. $m, n \geq N \Rightarrow d_Y(f_m(t), f_n(t)) \leq \epsilon$, $\forall t \in X$.

Therefore, for this N and $m, n \geq N$, $\lim_{t \rightarrow x} d_Y(f_m(t), f_n(t)) = d_Y(\lim_{t \rightarrow x} f_m(t), \lim_{t \rightarrow x} f_n(t)) \leq \epsilon$.

That is, $\{\lim_{t \rightarrow x} f_n(t)\}$ is a Cauchy sequence in Complete Y , it has a limit, say to A .

Meanwhile, since $f_n \rightarrow f$, we can choose $N_1 \in \mathbb{N}$ s.t. $n \geq N_1 \Rightarrow d_Y(f_n(t), f(t)) \leq \frac{\epsilon}{3}$, $\forall t \in X$.

And, choose $N_2 \in \mathbb{N}$ s.t. $n \geq N_2 \Rightarrow |\lim_{t \rightarrow x} f_n(t) - A| \leq \frac{\epsilon}{3}$.

And, since the limit exists, for this $n \geq N_2$, take $\delta > 0$ s.t.

$$t \in B_\delta(x) \setminus \{x\}, \quad |f(t) - \lim_{t \rightarrow x} f_n(t)| \leq \frac{\epsilon}{3}$$

Finally, $t \in B_\delta(x) \setminus \{x\}$ and $n \geq \max\{N_1, N_2\}$,

$$|f(t) - A| \leq |f(t) - f_n(t)| + |f_n(t) - \lim_{t \rightarrow x} f_n(t)| + |\lim_{t \rightarrow x} f_n(t) - A| \leq \epsilon.$$

□

Corollary. If $\{f_n: X \rightarrow Y\}$ is a sequence of continuous maps on a Metric space X into Complete Y , and $f_n \rightarrow f$, then f is continuous.

Uniform Convergence and Integration.

Thm. Let $\alpha: [a, b] \rightarrow \mathbb{R}$ be monotone increasing and $f_n \in \mathcal{B}(\alpha)$, $n \in \mathbb{N}$.

If $f_n \rightarrow f$ on $[a, b]$, then $f \in \mathcal{B}(\alpha)$ and

$$\lim_{n \rightarrow \infty} \int_a^b f_n d\alpha = \int_a^b \lim_{n \rightarrow \infty} f_n d\alpha = \int_a^b f d\alpha.$$

Proof. For each $n \in \mathbb{N}$, define $E_n = \sup_{x \in [a, b]} |f_n(x) - f(x)|$. Then, clearly

$$f_n - E_n \leq f \leq f_n + E_n \quad (\text{for all } t \in [a, b])$$

$$\text{Therefore, } \int_a^b f_n - E_n d\alpha \leq \int_a^b f d\alpha \leq \int_a^b f d\alpha \leq \int_a^b f_n + E_n d\alpha \quad (*)$$

$$\text{Hence } 0 \leq \int_a^b f d\alpha - \int_a^b f d\alpha \leq 2E_n \cdot [\alpha(b) - \alpha(a)] \rightarrow 0 \text{ as } n \rightarrow \infty.$$

Thus $f \in \mathcal{B}(\alpha)$. The equality is obtained by (*), using $f \in \mathcal{B}(\alpha)$.

Uniform Convergence and Differentiation.

Thm. Let $f_n: [a, b] \rightarrow \mathbb{R}$ be differentiable maps s.t. for some $x_0 \in [a, b]$, $\{f(x_0)\}$ converges.

If $\{f_n\}$ converges uniformly on $[a, b]$, then $\{f_n\}$ converges uniformly on $[a, b]$ to f with

$$\frac{d}{dx} \lim_{n \rightarrow \infty} f_n(x) = \lim_{n \rightarrow \infty} \frac{d}{dx} f_n(x).$$

Proof. Given $\varepsilon > 0$, take $N \in \mathbb{N}$ s.t. $n, m \geq N$ implies

$$|f_n(x_0) - f_m(x_0)| \leq \frac{\varepsilon}{2} \text{ and } |f'_n(t) - f'_m(t)| \leq \frac{\varepsilon}{2(b-a)}, \quad \forall t \in [a, b].$$

By the Mean-Value Theorem for $f_n - f_m$, for all $x, t \in [a, b]$,

$$|f_n(x) - f_m(x) - (f_n(t) - f_m(t))| \leq \frac{\varepsilon}{2(b-a)} \cdot (b-a) = \frac{\varepsilon}{2}. \quad (*)$$

Therefore, for any $x, t \in [a, b]$ and $n, m \geq N$,

$$|f_n(x) - f_m(x)| \leq |f_n(x) - f_m(x) - f_n(x_0) + f_m(x_0)| + |f_n(x_0) - f_m(x_0)| \leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

so that $\{f_n\}$ converges uniformly, let $f = \lim_{n \rightarrow \infty} f_n$. For each $x \in [a, b]$, define on $[a, b] \setminus \{x\}$

$$\phi_n(t) = \frac{f_n(t) - f_n(x)}{t - x} \quad \text{and} \quad \phi(t) = \frac{f(t) - f(x)}{t - x}$$

(*) gives ϕ_n is Cauchy sequence, thus converges uniformly, to ϕ . Finally,

the fact that ϕ_n continuous gives the result.

Summary. Let $f_n: X \rightarrow Y$ be functions, $f_n \rightarrow f$.

Continuity If f_n are continuous and $f_n \rightarrow f$, then f is continuous.

Differentiation If f_n are differentiable and $f_n \rightarrow f'$ and $f_n(x_0) \rightarrow f(x_0)$, then $f_n \rightarrow f$ and

$$\frac{d}{dx} \lim_{n \rightarrow \infty} f_n(x) = \lim_{n \rightarrow \infty} \frac{d}{dx} f_n(x)$$

Integral If $f_n \in \mathcal{B}(a)$ and $f_n \rightarrow f$, then $f \in \mathcal{B}(a)$ and

$$\int_a^b \lim_{n \rightarrow \infty} f_n \, dx = \lim_{n \rightarrow \infty} \int_a^b f_n \, dx$$